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WALTER SYDNEY ADAMS
1876–1956

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After a long and remarkably vigorous life Dr. Walter S. Adams passed away peacefully at his home in Pasadena on May 11, 1956. He was stricken with illness a few weeks previously and his strength gradually failed until the end. Since his retirement as Director of the Mount Wilson Observatory of the Carnegie Institution of Washington in 1946 he had continued his research at the Hale Laboratory. On December 20 he would have reached the age of eighty years. A memorial service was held at the library of the observatory on May 14.

Dr. Adams was born in the mountain village of Kessab near Antioch in northern Syria where his parents were serving as missionaries. Although the family returned to America a few years later, he imbibed some of the spirit of the Near East and took much pride in some of his earliest recollections. A few Turkish words for food and domestic articles remained with him.

His parents, Dora Francis Adams and Lucien Harper Adams, were of old Puritan stock from the state of New Hampshire, to which they returned after their foreign service. In the invigorating atmosphere of the "Granite State" Walter and his brother Edward took many adventurous trips on foot and acquired fixed ideas of personal independence and freedom as well as wiry sturdy physiques.

As a boy Walter attended Dartmouth College after taking his preparatory work at St. Johnsbury and Andover academies. In college he had a reputation as a brilliant student. His interest

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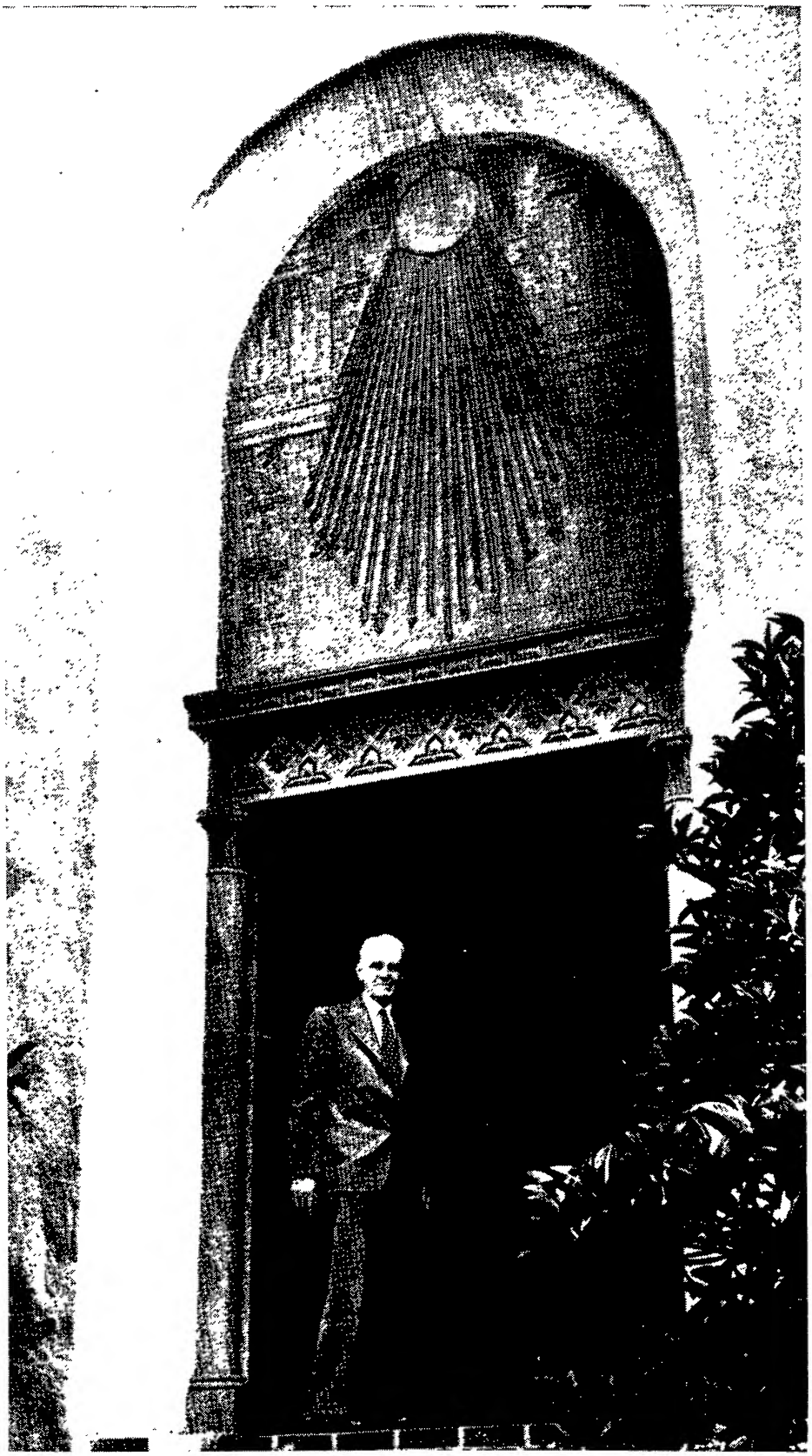

Dr. Walter S. Adams at the Hale Laboratory

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was turned to astronomy through the teaching of Professor Edwin B. Frost. In 1897 Frost, who had become widely known for his translation and revision of Scheiner's *Astronomical Spectroscopy*, was invited by Dr. George E. Hale to head the department of stellar spectroscopy at the newly established Yerkes Observatory. After graduation Adams followed his mentor westward, continuing his studies at the University of Chicago and taking up observational methods at the observatory where Frost was building a stellar spectrograph for use with the great 40-inch refractor. In 1900–1901 Adams spent a year in Munich, studying under the eminent German astronomers H. von Seeliger and K. Schwarzschild.

Returning to Yerkes in 1901 he worked until 1904 with Frost on the program of radial velocities of early-type stars, using the 40-inch refractor. Thus, he became interested in the possibilities of the spectrograph for studies of the stars—an interest which remained the dominant factor in his life for more than fifty years.

When the Carnegie Institution of Washington decided to support the new astronomical station on Mount Wilson, Dr. Hale chose from the Yerkes Observatory four young members of the staff to take part with him in the new establishment, W. S. Adams, F. Ellerman, G. W. Ritchey, and F. G. Pease. Of these four, Adams was the only one who had had systematic scientific training, and it was natural that he soon became the leader in the research programs under Hale's inspired planning. He enjoyed the early days on the mountain, which reminded him of those of his youth in New Hampshire and on the slopes of Mount Casius in Syria. In 1913 he was appointed assistant director and in 1923 he succeeded Dr. Hale as director.

In his characteristic way of making the best use of circumstances in which he found himself, Adams started with the equipment at hand on Mount Wilson although he had not previously taken part in solar research. He photographed the spectra of sunspots with the Snow telescope and worked with Hale on differences in the spectrum of the center and limb of the sun. He also laid out an extensive project of determining the rotation of the sun by measuring the Doppler displacements at the limb in different latitudes. At the same time, in high-dispersion spectra of

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α Boötis (4.3 Å/mm with the first order of the grating) and of α Orionis (10 Å/mm at λ 5400 with a 64° prism), it was found that those lines which were stronger in spots than on the disk of the sun were also strengthened in about the same amount in the spectrum of α Boötis and by even greater amounts in α Orionis. Lines which were weaker in spots were also weaker in the stars. These results, confirming the suspicion that the differences were due to differences in temperature, were supported by color differences in the continuous spectrum, and later fully tested and calibrated in the laboratory by using the furnace and different regions of the arc as sources.

These studies, which showed Adams' continued interest in stellar spectra, were of the highest importance to astronomy, not only for their contribution to solar physics, but because they led directly through the analysis of individual lines to reliable temperature and luminosity classification as well as to innumerable other results based on detailed studies of spectra.

This method is applicable to all stars which have suitable absorption lines sensitive to temperature and density in stellar atmospheres. In its operation it is entirely independent of distance. The intensities of certain lines were soon found to be correlated with luminosity or absolute magnitude, which, in turn, could be reduced to distance by comparing with the apparent magnitude. The absolute magnitudes used for calibration of the luminosity scale were deduced from trigonometric parallaxes or from proper motions, largely of near-by stars. Except for the unfortunate circumstance that the apparent magnitudes may need correction for interstellar absorption the spectroscopic distances based on apparent and absolute magnitudes have a high percentage accuracy at great distances, just where other methods become impossible.

Some of the lines sensitive to temperature and luminosity were originally noted by Adams in sunspots and in the spectrum of α Boötis. In the years 1913–14 A. Kohlschütter joined the search for additional sensitive lines by examining 60-inch reflector spectrograms of stars having trigonometric parallaxes. Pressure for more stellar parallaxes came from J. C. Kapteyn during his summer visits to Mount Wilson in those years, because he needed

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distances in his studies of star streaming. Later, relative line intensities were used, comparing the strength of a sensitive line with that of near-by invariable lines of the same spectrogram. The calibration was improved from time to time by the use of more material, especially by including the statistical computations of G. Stromberg based on proper motions. The method was also extended to a larger range of spectral types.

By 1935, the spectral types and absolute magnitudes of several thousands of stars had been estimated with sufficient precision for statistical work by Adams, Joy, and Humason at Mount Wilson.

In a paper entitled "Sunspots and Stellar Distances" (*Pub. A.S.P.*, **51,** 133, 1939) the fascinating story of this most fruitful research is told by Dr. Adams himself. It is clear that this project appealed to him as the best example in his own experience of the "very great intellectual satisfaction" which comes to an investigator in the field of physical science. It brought in large measure "the pure joy of discovery" to those who took part in it and provided boundless "stimulating and pleasing uncertainty" as to its usefulness in future developments of studies of the structure of the galaxy and the physical behavior of stars. All astronomers today would agree with him on the value of this method which adds a new parameter to the observable data on distant stars. The products of its use are essential to various attacks on the life history of stars and galactic systems.

Most of the spectrograms used in determining spectroscopic parallaxes were originally obtained for radial velocity measurement. The large velocity program was originally planned by Adams when spectroscopic observations began in 1911 at the 60-inch reflector. It was enlarged from time to time and extended to fainter stars with the 100-inch reflector. All members of the staff of the spectrographic department had a part in it. More than 7000 stellar velocities were determined in a 40-year period.

In this rather tedious work Adams took a never failing interest and contributed more than his share in the actual observing and measuring. The results were used in various investigations. By combining the proper motions, distances, and radial velocities, the space motions of large numbers of stars are now available. Dur-

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ing the years, great improvements in the instrumental equipment for spectrographic observation became possible and the development work was usually undertaken by Adams. He had a keen appreciation of the practical value of new equipment and seldom made unsuccessful changes.

Perhaps as a result of Hale's fervor for observing, Adams maintained from the beginning that the Mount Wilson Observatory was essentially an observing station. This policy was consistently maintained in his own work and in his administration.

In addition to the major programs of radial velocity, spectral classification, and stellar distances that involved thousands of stars and several decades in time, Adams took part in scores of investigations more limited in scope but of importance for our understanding of stars and their atmospheres. Most of these investigations are so well known that a mere reference to them here will suffice.

Among his most significant studies in early days should be mentioned the evaluation of the rotational velocity at the sun's limb in different latitudes with the Snow telescope and the 60-foot solar tower (1906–8), the effect of pressure on spectral lines in the laboratory (with H. G. Gale, 1912), the flash spectrum without an eclipse (1909–15), and the intensity of the continuous spectra of stars differing in proper motion and absolute magnitude or distance (1914).

Later, came the important and most difficult measures of the relativity displacement in the spectrum of the white dwarf companion of Sirius which definitely checked the Einstein prediction (1925), the calibration of Rowland's scale of intensities of lines in the solar spectrum (with H. N. Russell, 1927–28), and the estimates of the amount of water vapor and oxygen in the atmosphere of Mars together with the identification of carbon dioxide in Venus (with C. E. St. John and T. Dunham, 1925–34).

In more recent years Adams undertook investigations of the details of stellar spectra made possible by improvements in the coudé spectrograph of the 100-inch telescope. The dispersions used were 2.9, 4.5, and 10 Å/mm. Among the studies were: the relative displacements of lines of neutral and ionized atoms in different stars (1930), a comparison of the velocities given by differ-

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ent elements (1935) and by lines in the same star having different excitation potentials (1941), an extraordinarily valuable investigation of the identity of the elements in space which form interstellar absorption lines in the spectra of O and B stars with measures of the velocities in the line-of-sight of the clouds in which these lines have their source (1935–49), and finally a detailed examination of the spectrum of α Orionis which revealed a surrounding shell together with indications of remarkable activity in the lower atmosphere of this cool supergiant star (1937–56). The number of the *Astrophysical Journal* carrying the report of this last investigation was received in Pasadena during the final illness of its author.

In less extensive papers he reported on the spectra of many individual stars. He found that lines of the elements Zn, Sr, Si, and Na are especially strong in the spectra of certain stars; that many stars of types later than G5, both giants and dwarfs, have bright H and K lines whose width seems to be related to the absolute magnitude of the star; and that hydrogen emission is present in some M-type dwarfs of the lowest luminosity. Numerous spectroscopic binaries were found and the spectra of the components of visual pairs often introduced interesting problems. The study of variable-star spectra was begun as early as 1915 (T Tauri) and extensive studies of novae were made at the time of outbursts.

In many of Adams' investigations other members of the staff took part. They were given generous recognition as joint authors in the final reports. In practically all of such projects Adams contributed more than his share of time and effort. If observations were needed or measures were to be made he always seemed to find time for them. He enjoyed it from beginning to end, and when the undertaking had reached its completion he was anxious to have it written up and sent on its way to the printer. The actual writing of the paper usually fell to his mighty pen (the largest fountain pen I have seen in actual use). He had the remarkable faculty of organizing the content of the paper in his mind with uncanny precision. I have seen him write at one sitting in longhand a complete report on a considerable investigation with such

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accuracy of wording that no later change or editorial corrections were necessary.

In addition to the large number of papers for astronomical publications, numbering more than 250, Adams prepared the *Annual Reports of the Director of the Mount Wilson Observatory* for the Yearbook of the Carnegie Institution of Washington for nearly thirty-five years. The average length was about thirty-five pages, analyzing in detail the work of the observatory for the year and giving a résumé of the research of each member of the staff for the year. Abbreviated accounts were also supplied to the Astronomical Society of the Pacific and the American Astronomical Society. These reports were prepared with great care and constitute a fair and unvarnished record of the work of the Mount Wilson Observatory for the use of others working in the field or those interested in the methods and history of science. To the best of my knowledge no member of the staff ever took exception to the description of his work recorded in the *Annual Reports*.

The recognized scientific accomplishments of the Mount Wilson Observatory and the harmonious cooperation of the group of scientists forming the staff are indisputable evidence for the success of its director as an administrator. He would have been the last to claim any especial talent or even the slightest interest in this field. Minor difficulties, particularly those involving personalities, worried him beyond measure. The methods of Hale and Adams were far apart in these matters. Hale carried everything before him by sheer weight of his enthusiasm for the ends sought; Adams reached the same goal by letting matters take their course and interfering only where absolutely necessary in order to keep the peace. Both allowed great freedom to staff members and showed the greatest interest in the work of each.

In general operation Dr. Adams was often in a difficult position because he represented the staff on the one hand and the Carnegie Institution on the other, but with his loyalty to both parties a solution was readily arranged. His practice of Yankee frugality was highly thought of by the Trustees of the Institution but sometimes was not popular with the staff, especially if it affected requests for some pet instrument or for salary increases.

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Dr. Adams felt that much of the success of the observatory was due to the lack of friction among members of the staff. For this reason he gave high weight to the personality of those applying for positions at the observatory, no matter in what capacity. In general, his administration was conservative. He was likely to favor enlargement and improvement of existing programs which had proved successful rather than a venture into new fields which would require a large expenditure of time and funds for development. This attitude was partly because he did not think it good policy to use the regular current grants of the Institution for uncertain projects. If special funds were provided, as for Michelson's experiments for measuring the velocity of light and the diameter of stars, he was ready to give cordial support.

In the construction of the 200-inch telescope Adams took a leading but unassuming part. His advice was invaluable in developing the plans and in organizing the operation.

The normal time for Dr. Adams' retirement fell during the war years but the Carnegie Institution Trustees asked him to continue as director of the observatory in order to supervise several contracts for the federal government. He gave most of his time to these projects and encouraged participation by staff members. However, he did not enjoy the red tape and lost motion of government methods, so that the strain of these years took a heavy toll of his vitality.

As an observer with the spectrograph Dr. Adams was at his best. A well-exposed spectrogram of an unusual object was his greatest delight and he often spoke of his observing trips to Mount Wilson as "the only fun I have." Of the astronomers with whom I have worked no other except E. E. Barnard took so much pleasure in actual observing at the telescope. Adams was a superb observer and made the most of the time available to him. Although not by nature mechanically inclined he had, by intense application, trained himself until his adjustments of optical instruments were well-nigh perfect. If the weather was not favorable for good results he was able to take his misfortune with good grace whereas others, Barnard among them, would be thrown into the depths of despair.

Dr. Adams enjoyed his contacts with astronomers throughout

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the world and his advice was frequently sought and generously given in connection with new enterprises. Often he was able to supply material of value which had been in use at Mount Wilson. During the uncertain period of the war, from 1941 to 1948, he was asked by Sir Arthur Eddington to serve as Acting Executive Secretary of the International Astronomical Union and as a result the organization was maintained intact although no general meetings were possible.

Following his retirement on January 1, 1946, Adams continued his activities at the Hale Solar Laboratory in Pasadena, measuring and studying high-dispersion spectrograms obtained at the coudé focus of the 100-inch reflector. These studies of the last decade were among his most important contributions. During this period he also wrote a valuable paper on the history of the International Astronomical Union and in a popular vein two articles on the founding and early days of the Mount Wilson Observatory. For the years 1946–49 he was appointed an Associate of the Carnegie Institution of Washington.

While Dr. Adams' concern was centered almost exclusively in the Mount Wilson Observatory he took a lively interest in politics of the day and had very definite opinions regarding national policies and personalities, some of which suffered change with passing years. He considered himself a liberal, but he was a liberal of the old school. In action he was usually on the conservative side. Any concessions to the philosophy of socialism or fascism were to him unthinkable. Through the pages of the *New York Times*, which he read faithfully on the Pasadena streets while walking from his home to the observatory office, he kept himself well informed on world affairs.

In local affairs Dr. Adams took much interest although his spare time was insufficient to allow much participation. For forty years he served as member, part of the time as Chairman, of the Pasadena City Library Board and had much satisfaction in the outstanding showing made by the Library.

Dr. Adams was the recipient of the highest honors of the world of science. He was elected to the National Academy of Sciences (1917), was Draper Medalist, and Chairman of the Astronomy Section; he was elected to the Royal Swedish Academy

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and to the Royal Academy of Sciences of Uppsala; he was a Foreign Member of the Royal Society of London, and Janssen Medalist of the Paris Academy of Sciences.

Honorary degrees were conferred on him by Dartmouth, Columbia, Pomona, Southern California, Chicago, and Princeton.

He was for fifty years a Fellow of the Royal Astronomical Society, an Associate, and Gold Medalist. In the International Astronomical Union he served as Vice-President, Chairman of the Commission of Stellar Radial Velocities, and Acting Secretary. He was a member of the American Astronomical Society (President); Astronomical Society of the Pacific (President, Bruce Medalist); American Association for the Advancement of Science (Vice-President and Chairman of Section D, President of Pacific Division); and the American Philosophical Society.

As a part of his heritage, Dr. Adams had a deep sense of the dignity and meaning of life, despising shallowness and pretense. At first meeting he seemed reserved and austere but this impression vanished with better acquaintance. Social affairs were avoided but friendly contacts with others in small groups gave him great pleasure and his fund of anecdotes concerning people he had known always seemed to enliven the occasion. His favorite pastimes included detective stories or a game of bridge, while the lure of golf or tennis would often induce him to leave his desk or measuring machine for an hour or two.

In 1910 Dr. Adams was married to Lillian Wickham, who died in 1920. In 1922 he was married to Adeline L. Miller who survives him, together with their two sons, Edmund M. and John F., and two grandchildren.

In the years to come, many of the researches of Dr. Adams, especially the method which he developed for determining the spectral classes and luminosity of stars from slit spectrograms, will be of fundamental value to astronomers. To his friends and colleagues his kindliness, good cheer, and devotion to duty will be a happy memory and continual source of inspiration.

For astronomy in general, the passing of Walter Sydney Adams marks the end of an era in the history of astronomy—an era of pioneer exploration in the distant expanse of our galaxy and the universe. In this period the accumulation of data by the

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best available instrumental means was all-important. At the present time it is difficult for us to realize that in the course of a single generation most of our basic knowledge of the size, content, structure, and dynamics of the astronomical world has been gained. To this great and inspiring enterprise Dr. Adams gave his life with zealous abandon and made unequaled contributions both through his own activities and through personal as well as official encouragement to many others.

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